

1. Définition et principes de la méthode Agile

The main idea

Agile = build the project progressively, get feedback frequently, and adapt to change.

Imagine you're building a website for a client.

Instead of saying:

"Let's spend 6 months designing everything, then build everything, and finally show it to the client."

Agile says:

"Let's build a small useful part, show it to the client, get feedback, improve it, then continue."

So Agile is based on:

Plan → Build → Test → Show → Get feedback → Adapt → Repeat

The important principles

You don't need to memorize all the Agile principles initially. Understand these:

- **Customer collaboration** → work closely with the client/user.
- **Working software** → prioritize something that actually works.
- **Responding to change** → requirements can change, and that's normal.
- **Continuous improvement** → regularly ask: *How can we work better?*
- **Frequent delivery** → deliver small pieces instead of waiting until the very end.

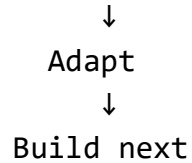
Mental model

Traditional thinking:

Plan everything → Build everything → Deliver

Agile thinking:

Plan → Build small → Deliver → Feedback



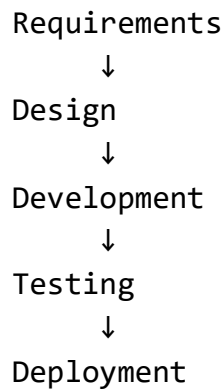
2. Méthode Agile vs méthode en cascade

Cascade = Waterfall.

The fundamental difference is **how they deal with change**.

Waterfall

You generally go through predefined phases:



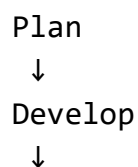
You finish one major phase before moving to the next.

The assumption is essentially:

"We can understand and define most of the requirements before development."

Agile

The phases happen repeatedly in smaller cycles:



Test



Feedback



Improve



The assumption is:

"We will discover and refine some requirements while building the product."

Simple example

A client says:

"I want an e-commerce website."

With Waterfall, you try to define **everything** at the beginning:

- all pages
- all features
- database
- design
- payment system
- reports
- etc.

Then development starts.

With Agile, you might first build:

Product catalog → show it → get feedback

Then:

Shopping cart → show it → get feedback

Then:

Authentication → show it → get feedback

Then:

Payment → show it → get feedback

So:

Waterfall	Agile
Plan heavily upfront	Plan progressively
Sequential phases	Iterative cycles
Change is expensive	Change is expected
Feedback often comes late	Feedback comes frequently
Delivery usually at the end	Frequent deliveries
Requirements relatively stable	Requirements can evolve

Important: Agile doesn't mean "**no planning.**"

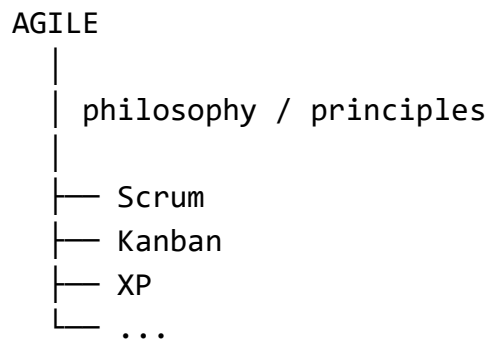
It means **planning is continuous and adaptable.**

3. Scrum comme un framework Agile

This is a very important distinction:

Agile is the philosophy/approach. Scrum is a framework used to apply Agile principles.

Think:

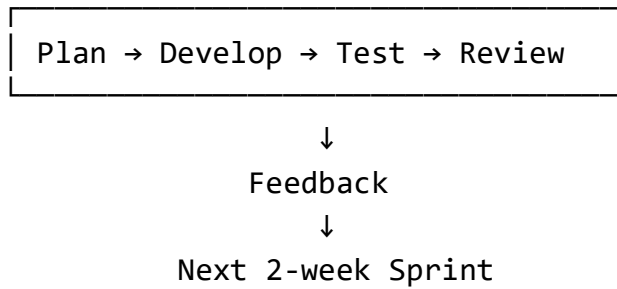


Scrum

Scrum organizes work into short periods called **Sprints**.

For example:

2-week Sprint



During each Sprint, the team selects some work and tries to produce a **usable increment of the product**.

Scrum has some important concepts

Product Backlog

The list of things the product needs.

Product Backlog

- |— Login
- |— Registration
- |— Product catalog
- |— Shopping cart
- |— Payment
- |— Reports

Sprint

A short development cycle, often 1–4 weeks.

Daily Scrum

A short daily meeting where the team synchronizes.

Sprint Review

The team looks at what was produced and gets feedback.

Sprint Retrospective

The team asks:

What went well?

What didn't go well?

How can we improve?

And then the next Sprint begins.

4. Avantages et limites

Advantages of Agile

1. Adaptability

If the client changes their mind:

"Actually, we don't need this feature anymore."

Agile can adapt relatively easily.

2. Frequent feedback

You don't wait six months to discover:

"This isn't what the customer wanted."

You discover problems earlier.

3. Early delivery

The customer can start receiving useful functionality before the entire project is finished.

4. Continuous improvement

The team doesn't only improve the product; it also improves **how it works**.

5. Better visibility

The customer and team can regularly see the real state of the project.

Limitations of Agile

Agile isn't magic.

1. Requires active collaboration

If the customer never gives feedback, Agile becomes difficult.

2. Requires a mature team

The team needs communication, responsibility and the ability to make decisions.

3. Scope can evolve

Because requirements can change, it can sometimes be difficult to know exactly what the final product will look like.

4. Can be poorly implemented

Some organizations say:

"We're Agile!"

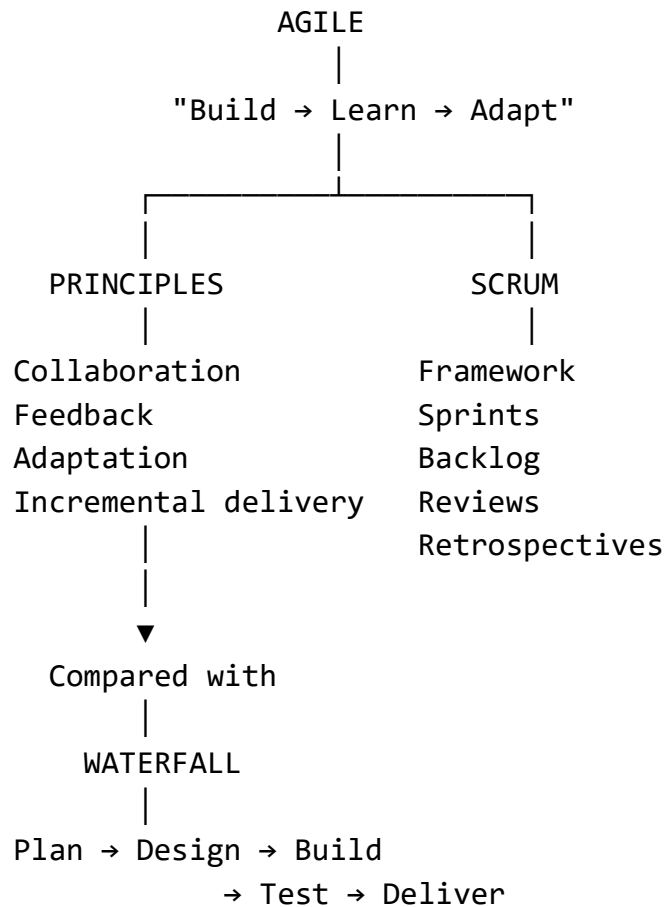
but simply have lots of meetings without actually applying Agile principles.

That's sometimes called "**Agile theater.**"

5. Not ideal for every project

For projects where requirements are extremely stable and predictable, a more traditional approach can sometimes make more sense.

The whole lesson in one picture



remember only 4 sentences:

1. **Agile is a philosophy based on iteration, feedback, collaboration and adaptation.**
2. **Waterfall tries to define and plan most things upfront and follows sequential phases.**
3. **Scrum is a framework that helps teams apply Agile through short Sprints and defined practices.**
4. **Agile's main strength is adapting to change; its main challenge is that it requires strong collaboration and discipline.**